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COMPE 470L

Lab 1 Report

Introduction

In this lab we refresh our memories of HDL design using Verilog by designing a one-digit Binary Coded Decimal Counter module and testbench.

Verilog Code

BCD Module

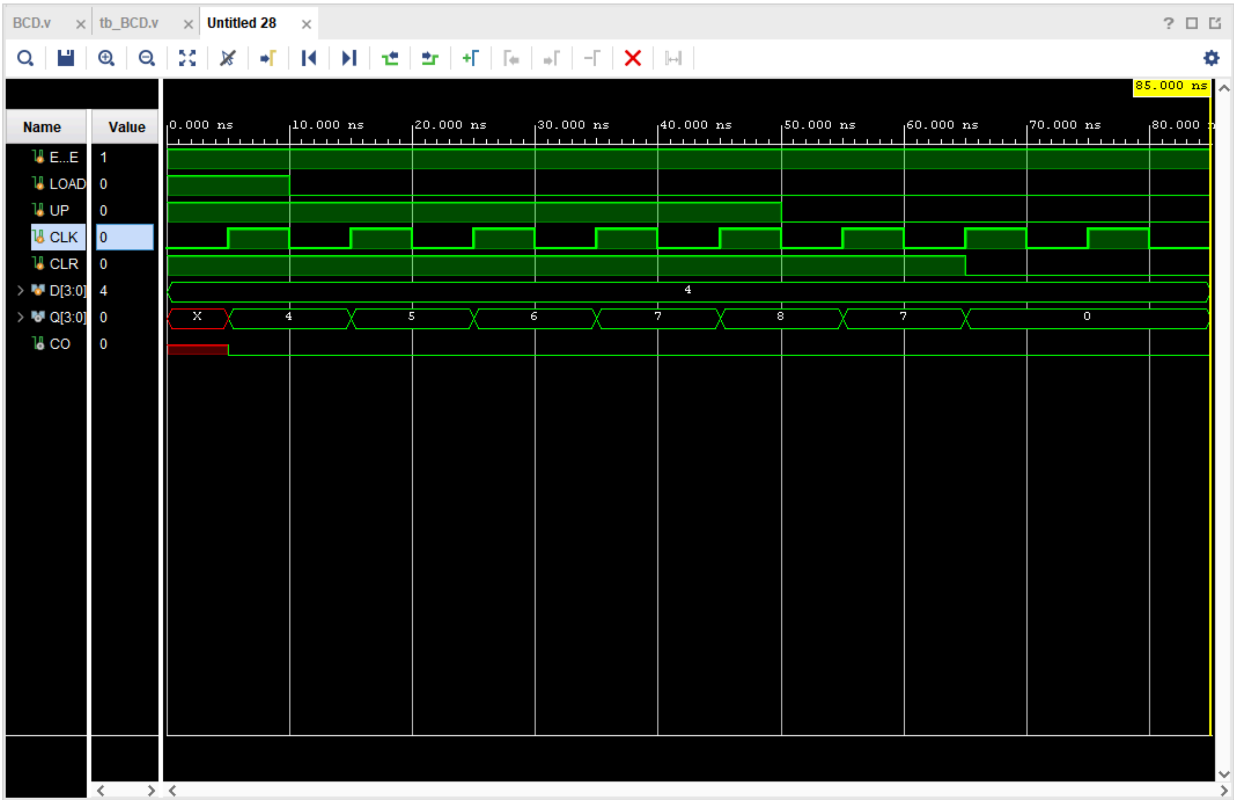
```
module BCD(input ENABLE, LOAD, UP, CLK, CLR,
    input [3:0] D,
    output reg [3:0] Q,
    output reg CO);

    always @(posedge CLK or negedge CLR) begin
        if (!CLR) begin     // ASYNC CLEAR
            Q <= 4'b0000;
            CO <= 1'b0;
        end
        else begin
            if (LOAD && ENABLE) begin // load
                CO <= 1'b0;
                Q <= D;
            end
            if (!LOAD && ENABLE && UP) begin // increment
                if (Q == 9) begin   // overflow
                    CO <= 1'b1;
                    Q <= 4'b0000;
                end
                else begin
                    CO <= 1'b0;
                    Q <= Q + 1;
                end
            end
            if (!LOAD && ENABLE && !UP) begin // decrement
```

```
            if (Q == 0) begin   // underflow
                CO <= 1'b1;
                Q <= 4'b1001;
            end
            else begin
                CO <= 1'b0;
                Q <= Q - 1;
            end
        end
    end
end
endmodule
```

BCD Testbench

```
`timescale 1ns / 1ps

module tb_BCD;

    reg ENABLE, LOAD, UP, CLK = 1'b0, CLR;

    always begin
        #5; CLK <= ~CLK;
    end
    reg [3:0] D;
    wire [3:0] Q;
    wire CO;

    BCD A(.ENABLE(ENABLE), .LOAD(LOAD), .UP(UP), .CLK(CLK), .CLR(CLR),
            .D(D), .Q(Q), .CO(CO));

    initial begin
    ENABLE = 1'b1; LOAD = 1'b1; UP = 1'b1; CLR = 1'b1;
    D = 4'b0100; #10; // load 4 into counter
    LOAD = 1'b0; #40;
    UP = 1'b0; #15;
    CLR = 1'b0;


    #20; $finish;
    end


endmodule
```

Results



Conclusion

In this lab I designed a one-digit BCD counter module and testbench. I designed it using a behavioral level description.